

Problem 1. Outlining Proofs

(a) **Theorem:** “Every natural number is tall or wide, but not both.”

Proof by cases. Suppose n is any natural number. We must show that it cannot be tall and wide at the same time. To show n must be either tall or wide assume n is not tall [TODO: prove that n is wide]. now all that remain is to show that n cannot be both tall and wide by cases.

Case I: n is tall. [TODO: show that n cannot be wide.]

Case II: n is wide. [TODO: show that n cannot be tall.]

Thus, by cases, we have shown that the theorem holds for every natural numbers. $\square$

(b) **Theorem:** “There exists a happy natural number between 10 and 1000 that has no happy proper divisors.”

Proof by example. Theorem holds for $n = 25$,

Show that 25 is happy natural number between 10 and 1000 because [TODO].

Also prove that 25 has no happy proper divisors [TODO]. $\square$

(c) **Theorem:** “There are no devious natural numbers.”

$$P(n) := n \text{ is not devious}$$

Proof by induction. **Induction hypothesis.** $P(n)$ holds for every $n \in \mathbb{N}$

Base Cases.

- $P(0)$ holds because [TODO].
- $P(1)$ holds because [TODO].

Inductive step: Assume $P(n - 1)$ holds for some arbitrary $n \in \mathbb{N}$, now to prove that $p(n)$ also holds because [TODO].

By induction principle, for every natural number $P(n)$ holds. $\square$

Problem 2. Spotting Errors

(a) Brynmor makes mistake while Plugging equations into equality. he forgot to flip the inequality when multiplying by -8 , the correct resultant equation would be $6x_i + 8x_{i-1} \geq 6 \cdot 2^{i-1} + 8 \cdot 2^{i-1}$ which is not sufficient to prove the inductive step.

(b) In given proof during inductive steps Erik stated that $x_i > 0$ which was yet to be proven.

(c) Zach’s inductive step fails on $i = 1$ because equation used in inductive step $x_{i+1} = 6x_i - 8x_{i-1}$ would require assumption that $S(0)$ to use value x_8 which is not defined.

(d)

Proof. We will use induction on stronger hypothesis $R(i) := "x_i \geq 3x_{i-1} - 1"$ AND $x_i > 0$, proving that $R(i)$ holds for $n \geq 2$.

Base Case: $R(2) = 13 > 12$, and $13 > 0$, base case holds.

Inductive step: Suppose for $n \geq 2$ $R(i)$ holds and let's show that $R(i+1)$ holds. Consider $x_{i+1} = 6x_i - 8x_{i-1}$ also from induction we know that $x_i \geq 3x_{i-1}$, So $\frac{8}{3}x_i \geq 8x_{i-1}$, now plug that into $x_{i+1} = 6x_i - 8x_{i-1} = \frac{10}{3}x_i + \frac{8}{3}x_i - 8x_{i-1}$ and you get $x_{i+1} \geq \frac{10}{3}x_i$. also $x_i > 0$ which means $\frac{10}{3}x_i > 3x_i$, $x_{i+1} \geq 3x_i$ and here $x_i > 0$ so, $x_{i+1} > 0$. That completes the Inductive step, so $R(i)$ is true for all $i \geq 2$.

$R(i)$ is true for $i \geq 2$ and $x_1 = 4$, so for all $i \geq 1$, $x_i > 0$ as desired. $\square$

Problem 3. Integer Multiplication from Bit Shifts

(a) start state to final state:

$$Q_1(5, 22, 0) \longrightarrow Q_2(10, 11, 0) \longrightarrow Q_3(20, 5, 10) \longrightarrow Q_4(40, 2, 30) \longrightarrow Q_5(80, 1, 30) \longrightarrow Q_f(160, 0, 110)$$

in final state $a_f = 110$ as desired, hence state machine computes $5 \cdot 22$ correctly.

(b) Theorem:- $P(r, s, a) := "rs + a = xy"$ is invariant.

Proof by invariant principle. : For start state Q_0 where $r = x, s = y, a = 0$ $P(x, y, 0)$ holds. Now to prove P is preserved we assume $P(r, s, a)$ holds (i.e: $rs + a = xy$) and that $(r, s, a) \rightarrow (r_t, s_t, a_t)$. We must prove that $P((r_t, s_t, a_t))$ holds, that is,

$$r_t s_t + a_t = xy$$

we also prove by cases on transition from $(r, s, a) \rightarrow (r_t, s_t, a_t)$:

• **Case s is even:**

$$(r_t, s_t, a_t) = (2r, s/2, a)$$

$$2r * s/2 + a = rs + a = xy$$

which is true by our assumption, so $P((r_t, s_t, a_t))$ holds.

• **Case s is odd:**

$$(r_t, s_t, a_t) = (2r, (s-1)/2, a+r)$$

$$2r * (s-1)/2 + a + r = rs + a = xy$$

is true by our assumption, so $P((r_t, s_t, a_t))$ holds.

That covers all the transitions and P is preserved, so by invariant principle $P(r, s, a)$ is invariant.

(c) state (r_f, s_f, a_f) is final when $s_f = 0$

(d) From part (b) for the final state $P((r_f, s_f, a_f)) = r_f s_f + a_f = xy$, also using (c) in the expression, we conclude that in any reachable final state $a_f = xy$.

(e) $bitcount(s) = \lceil \log_2(s+1) \rceil$ is derived variable so that $s \in q = (r, s, a)$ where $q \in Q$.

Now for some arbitrary transition $q \rightarrow q'$, by definition of transition, value of s is at least getting halved and so s_{new} is at most $s/2$ and we also know that halving the value of a positive integer would lead to its bit count being reduced by 1, meaning $bitcount(s) > bitcount(s_{new})$

By the given facts, we can conclude that $bitcount(s)$ is strictly decreasing and its output is strictly closed over natural numbers.

This implies that the algorithm will eventually terminate because $bitcount(s)$ is strictly decreasing by each transition and it has a non-negative lower bound, so it must terminate.

(f) According to Termination Theorem, since $bitcount(s)$ is strictly decreasing derived variable defined over set of natural numbers (By Part(e)) it will end in at most $\lceil \log_2(y+1) \rceil$ steps when starting state is $(x, y, 0)$. $\square$

Problem 4. Invariant Enlightenment

Definition: For our proof, perimeter is the total number of edges between enlightened and unenlightened students.

Lemma 4.1. $P(n) :=$ With k initially enlightened students, after n days the perimeter will be at most $4k$.

Proof by induction. :

Base Case $P(0)$: On day 0, if the k enlightened students are seated far apart, their perimeter will be at most $4k$, base case holds.

Inductive step: Assume $P(n)$ is true and now want to show that $P(n+1)$ is also true. On day $n+1$, new students can become enlightened if 2 or more students adjacent to them are enlightened, so when a new student is enlightened, we observe Net change in perimeter by cases on the number of adjacent students.

- **Case I (2 adjacent):** increase and decrease by 2.
- **Case II (3 adjacent):** decrease 3 and increase by 1.
- **Case III (4 adjacent):** only decrease by 4.

All cases by which the perimeter may change are exhausted, and in all of the cases we showed that the total perimeter will stay $4k$ at most, implying $P(n+1)$ and By induction hypothesis $P(n)$. $\square$

Theorem 4.2. *If fewer than n students among those in an $n \times n$ arrangement are initially enlightened, and additional students reach enlightenment only by being taught by two or more neighbors, then there will be at least one student who never reaches enlightenment.*

Proof. Let's say that k students were initially enlightened in some $n \times n$ grid, by the end of all possible transitions, every student in the grid is enlightened. From Lemma 4.1, we know that the perimeter of such a state can be $4k$ at most, we also know that perimeter of $n \times n$ grid is $4n$.

$$4k \geq 4n \implies k \geq n$$

Which means that to completely enlightened grid of $n \times n$ students initially we need at least n students. Which proves the theorem. $\square$